Derivation of spectrally resolved flux at the top of the atmosphere from EMIT hyperspectral observations:methodology and validation with CERES

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Abstract

1. Introduction

Dust aerosols play a crucial role in the climate system due to their significant radiative effects, both direct and indirect. Directly, they influence the Earth's radiation balance by scattering and absorbing solar and terrestrial radiations, which can either cool or warm the atmosphere depending on the properties of dust and environmental factors such as surface reflectance (Myhre et al., 2003; Song et al., 2022; Yu et al., 2006; Zhang et al., 2016). This interaction affects surface temperatures and stability of the lower atmosphere and therefore may alter weather patterns (Bretl et al., 2015; Miller and Tegen, 1998; Pérez et al., 2006; Reale et al., 2014). Indirectly, dust aerosols act as cloud condensation nuclei and ice nuclei, influencing cloud formation, microphysical properties and lifetime, which in turn affect precipitation patterns and cloud albedo (DeMott et al., 2003; Gu et al., 2012; Kanji et al., 2017). Additionally, dust aerosols significantly impact air quality, contributing to particulate matter pollution that harms human health and reduces visibility (Chin et al., 2007; Longueville et al., 2010; Wang et al., 2020). Understanding the complex role of dust aerosols is essential for improving climate modeling and assessing their broader environmental and health implications.

Substantial progress has been made over the last few decades in quantifying the DRE of dust aerosols using Earth system models (ESMs), but key uncertainties persist. The availability and improvement of satellite observations of dust aerosols, particularly aerosol optical depth (AOD), have provided invaluable data for evaluating and refining **s**imulations of dust loading and DRE in ESMs (references). However, DRE is influenced not only by AOD but also by other dust properties, such as particle size and mineralogical composition. Unfortunately, there has been **a** general lack of observations for these properties, leading to significant uncertainties in model-based estimations of dust DRE.

For example, multiple lines of evidence suggest that ESMs underestimate dust particle sizes, leading to an overestimation of the cooling effect of dust DRE, though the mechanisms sustaining super-coarse mode dust in the atmosphere remain poorly understood **(Biagio et al., 2020; Kok et al., 2017; Ryder et al., 2019; Zheng et al., 2024)**. Additionally, the mineral composition of dust particles, which is linked to the soil mineralogy of their source regions, strongly influences their solar radiation absorption. Until recently, observations of both soil and airborne dust mineralogy have been scarce. Consequently, ESMs often assume a fixed dust mineral composition and corresponding refractive index in DRE estimations. In this regard, observation-based studies are essential for reducing uncertainties in dust DRE estimations and enhancing ESM performance.

Two different methods are frequently used in previous studies to derive dust DRE from observations.The first method involves using retrieved dust properties, particularly AOD, to drive radiative transfer (RT) models for computing dust DRE (hereafter referred to as the "RT-based method") (Hansell et al., 2010; Kuwano et al., 2024; Song et al., 2022). The RT-based method offers high flexibility, as it can estimate DRE at both the top of the atmosphere (TOA) and the surface and can be applied across different temporal scales, from instantaneous snapshots to diurnal or monthly means. However, a key limitation of this approach is that many dust properties other than AOD must often be assumed due to a lack of observational constraints, introducing significant uncertainties in RT simulations (Song et al., 2018). The second commonly used method is to diagnose dust DRE solely based on data analytics without using RT simulations (referred to as "data-driven method"). Take the SW dust DRE at the TOA for example. It is defined as,

$$DRE_{TOA} = F^{\uparrow}_{clear} - F^{\uparrow}_{dust} \quad , \tag{1}$$

Where $F^{\uparrow}_{clear}$ and $F^{\uparrow}_{dust}$ represent the reflected SW radiation at the TOA for dust-free and dust-laden atmospheres, respectively. At a given location, $F^{\uparrow}_{clear}$ can be estimated from observations, such as CERES, during periods of minimal dust loading (e.g., very low AOD), while $F^{\uparrow}_{dust}$ is derived from observations under significant dust loading. Alternatively, some studies estimate DRE_{TOA} through regression of radiation observations with collocated AOD retrievals (Kuwano et al., 2024; Loeb et al., 2021; Song et al., 2018). A key advantage of the data-driven method is that it does not require explicit assumptions about dust properties, such as particle size and refractive index, which are difficult to measure. However, it is less flexible than the RT-based method. For example, if $F^{\uparrow}_{clear}$ and $F^{\uparrow}_{dust}$ are instantaneous observations at the TOA, extending the DRE results to other temporal scales (e.g., diurnal or monthly means) or to the surface is challenging.

Regardless of the methods used, CERES broadband flux products, which are well calibrated and extensively assessed, are frequently used in the studies of dust DREs. However, because of its broadband nature, CERES products lack spectral information which can be especially useful for constraining ESM simulations. For example, several studies have shown that the spectrally-resolved outgoing longwave radiation (OLR) poses a much stronger constraint on the water vapor and cloud simulations in the ESMs (Huang et al., 2024, 2008, 2010, 2013). The spectrally-resolved OLR in these studies were often derived from the hyperspectral LW radiance observations from sensors like Atmospheric Infrared Sounder (AIRS) collocated with the reference CERES observations (Huang et al., 2008, 2010). These studies clearly show that the spectrally-resolved flux contains more information content than the broadband CERES observation, which is inspiring for the SW dust DRE studies. For example, (Song et al., 2018) showed that the use of smaller dust particle size and more absorptive refractive index combination can lead to almost identical dust DRE_{TOA} simulations as the combination of larger size and less absorptive refractive index. In other words, using the CERES broadband flux observations alone would not be able to

distinguish the difference between the two combinations. Conversely, a spectrally-resolved flux observation may provide an important spectral signature of DRE_{TOA} that can be used to delineate the impacts of dust size and refractive index.

In 2022, the Earth Surface Mineral Dust Source Investigation (EMIT) mission, developed by NASA Jet Propulsion Laboratory (JPL), was launched to the international space station (Green et al., 2020). The primary objective of the EMIT mission is to retrieve the mineralogical composition of surface soils, which can then be employed to improve the simulation of the mineral composition of airborne dust in ESMs. The EMIT instrument provides hyperspectral observations of reflected solar radiation from Earth-atmosphere systems over major desert regions, spanning wavelengths from approximately 380 nm to 2500 nm, at a spectral resolution of 7.4 nm (see details in Section 2.1). In addition to EMIT, several other satellite instruments launched recently, including the Ocean Color Instrument (OCI) on the PACE mission and the TROPOspheric Monitoring Instrument (TROPOMI) on the Sentinel-5 Precursor, also provide hyperspectral SW reflection observations. The broad spectral coverage and high spectral resolution of EMIT offer an unprecedented opportunity to observe the spectral signatures associated with dust reflection. As demonstrated by previous studies in the longwave (LW) region, spectrally resolved fluxes provide a stronger constraint on ESMs than broadband integrated fluxes. Furthermore, a spectrally resolved flux product derived from EMIT observations can be employed either to directly estimate dust DRE using data-driven methods, or to serve as a benchmark for evaluating DRE derived from RT-based approaches.

Mention previous studies of deriving flux form say MODIS. But these studies used CERES as the regression. Our method uses the EMIT among without regression with CERES.

This paper documents the method we developed to derive the spectrally-resolved fluxes at the TOA from the EMIT hyperspectral observations in the Sahara region. We will first provide an overview of the data

that were used in this study, including the EMIT, PACE-OCI and CERES products (Section 2). Then we will explain the methodology used to derive the spectrally-resolved fluxes at the TOA from the EMIT and show the evaluation of the results through comparisons with the collocated CERES flux products (Section 3).

2. Data

In this study, we use EMIT, PACE-OCI and CERES products.

2.1. Hyperspectral Observations of EMIT

Launched to the ISS in 2022, the primary objective of the EMIT mission is to map the mineral composition of arid dust source regions, enabling improved representation of dust mineralogy and its associated DRE in ESMs. The EMIT imaging spectrometer collects measurements of reflected sunlight from the Earth–atmosphere system across 285 spectral bands, spanning wavelengths from 381 nm to 2493 nm, with a spectral resolution of 7.4 nm and a spatial resolution of 60 m. This broad spectral coverage allows EMIT to capture the majority of the incoming solar energy. As illustrated in Figure 1, the EMIT spectral range (381–2493 nm; red shaded region) accounts for 90.1% of the total solar irradiance at the top of the atmosphere (TOA). The remaining 9.9% consists of 6.6% from the ultraviolet (UV) region (<381 nm; purple shaded region) and 3.3% from the shortwave infrared (SWIR) region (>2493 nm; gray shaded region). It is important to note that this partitioning is based on the solar insolation spectrum. The distribution differs substantially for the reflected spectrum at the TOA, as it depends on surface reflectance and atmospheric composition, including gases, aerosols, and clouds. For instance, the energy contribution in the UVA region (<314 nm) is negligible due to strong ozone absorption,

which likely explains why the insensitivity of the spectral response function (SRF) of the CERES SW instrument in the UVA does not compromise its accurate TOA flux measurements (Loeb et al., 2016). Similarly, the energy in the SWIR region with wavelengths larger than 2493 nm is also negligible due to the absorption of water vapor. The only spectral region that has significant energy and yet is not captured by EMIT is the UVA region between 314 nm and 380 nm. As shown later, this part of the spectrum, which is missed by the EMIT, will be estimated from the spectrum of 381 nm to 440 nm, which is observed by EMIT, based on an empirical relation derived based on the PACE OCI observations. In summary, in this study the upward reflected SW flux at the TOA derived from the EMIT observations covers the spectral region from 315 nm to 2493 nm.

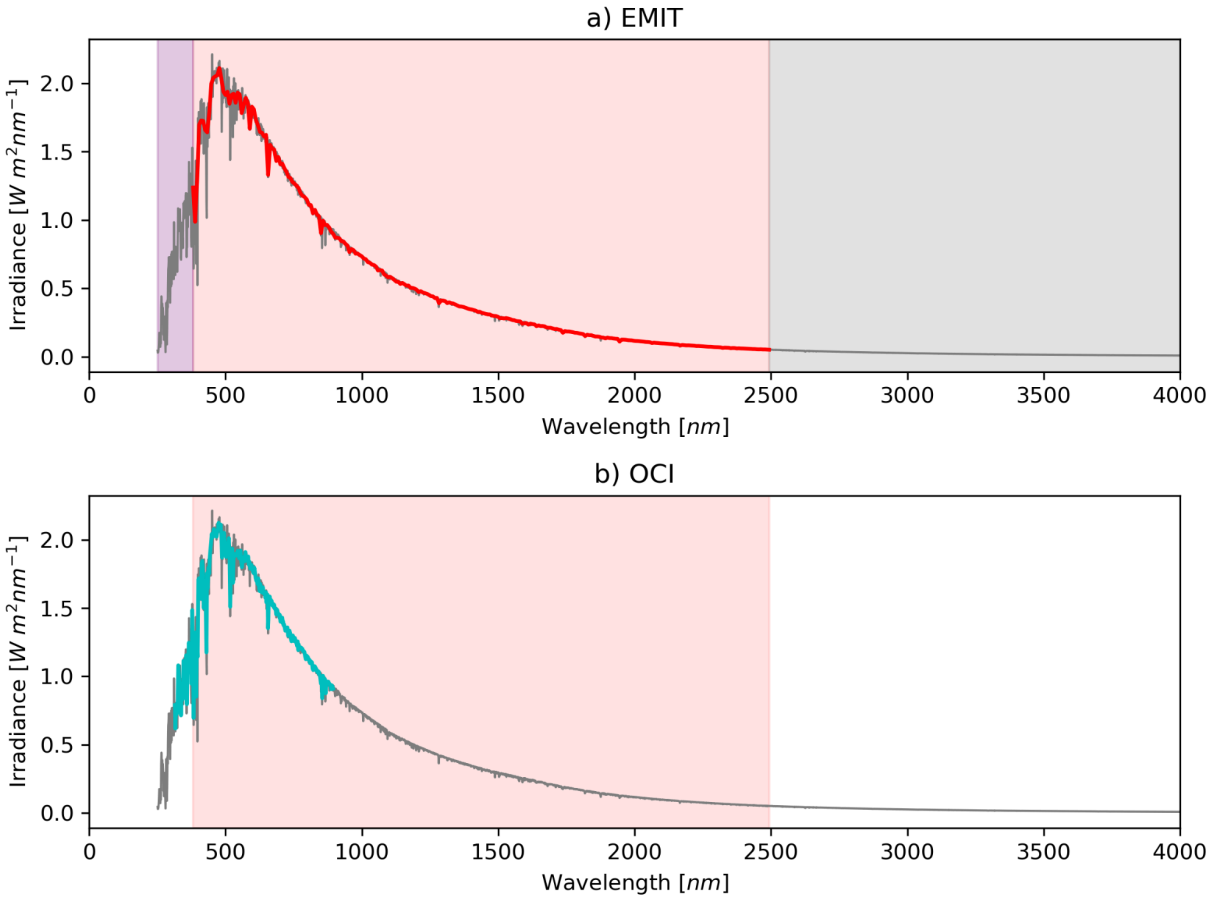


Figure 1

Spectral region	Wavelength range	
UVB and shorter	< 314 nm	Neglected
UVA	314 nm to 381 nm	Empirical regression
Visible to SWIR	381 nm to 2493 nm	EMIT observation
SWIR and longer	>2493 nm	Neglected

2.2. Hyperspectral Observations of PACE OCI

The Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission launched in early 2024 aims to extend and improve our understanding of ocean biology, aerosols, and clouds, ultimately benefiting our understanding of ocean health, air quality, and climate change (Werdell 2019). The PACE mission includes three science instruments, the primary hyperspectral Ocean Color Instrument (OCI) and two polarimeters. The OCI is a hyperspectral imaging radiometer built at Goddard Space Flight Center (Meister 2024). It provides contiguous spectral coverage extending from 315 nm in the UV to 895 nm in the near infrared (NIR) spectrum at 2.5 nm resolution (with bandwidths of 5 nm). In addition to the hyperspectral observations in the UV-NIR, OCI also provides reflectance observations in 7 discrete shortwave infrared (SWIR) bands extending the spectral coverage further to 2260 nm. The OCI telescope scans from west to east at a rotation rate of 5.77 Hz, acquiring Earth view data at a 1.2 km × 1.2 km ground sample footprint at the center of the scan (higher at the scan edges) and a ground swath width of ~2700 km.

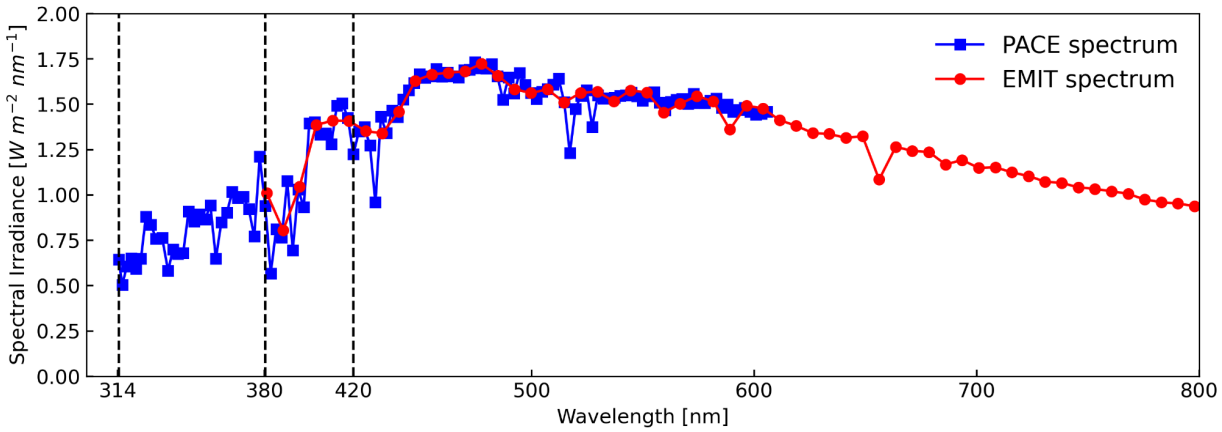


Figure 2.

The primary purpose of utilizing OCI observations in this study is to establish an empirical relationship between solar flux in the UVA region (315–380 nm) and the spectral region between 381 and 440 nm (referred to as the "UV-blue" region in this study). As illustrated in Figure 2, based on collocated observations, the OCI blue channel spectrum shows excellent agreement with EMIT for wavelengths longer than 380 nm, while providing additional spectral coverage in the UVA region. We propose that if an empirical relationship can be derived between the UVA and UV-blue regions from OCI observations, this relationship can be applied to the EMIT spectrum to estimate the missing UVA energy using EMIT's UV-blue observations. The methodology for deriving this empirical relation is detailed in Section 3.1, and the results are presented in Section 4.

2.3. CERES SSF product

The CERES (Clouds and the Earth's Radiant Energy System) sensor dedicated to monitoring Earth's energy budget is a three-channel scanning radiometer measuring the broadband outgoing shortwave (0.3-5 μm), window-region (8-12 μm) and longwave (5-100 μm) radiances at TOA with a spatial resolution of 20 km at nadir (Loeb et al., 2001; Priestley et al., 2011). Among various CERES data products, the Single Scanner Footprint (SSF) product merges instantaneous radiative flux observations with coincident cloud and aerosol retrievals, providing collocated and detailed characterization of cloud properties, aerosol conditions, and radiation fluxes at high spatial resolution. In this study, we use the SSF product from the CERES on board of the Terra satellite to evaluate the TOA SW flux derived from the EMIT observations.

3. Methodology

Theoretically, the spectral hemispherical flux $F(\lambda)$ can be derived from the observation of spectral radiance $I\left(\lambda, \theta_0, \theta, \phi\right)$ based on the following equation:

$$F(\lambda) = \frac{\pi I(\lambda,\theta_0,\theta,\phi)}{R(\lambda,\theta_0,\theta,\phi)}, \quad (1)$$

where λ indicates wavelength, θ_0, θ, and ϕ correspond to the solar zenith, viewing zenith and relative azimuth angles, respectively. $R(\lambda, \theta_0, \theta, \phi)$ in the denominator is the angular distribution model (ADM) anisotropic factors defined as:

$$R(\lambda, \theta_0, \theta, \phi) = \frac{\pi I(\lambda,\theta_0,\theta,\phi)}{\int_0^{2\pi}\int_0^{\pi/2} I(\lambda,\theta_0,\theta,\phi) cos\theta sin\theta d\theta d\phi}. \quad (2)$$

The value of $R(\lambda, \theta_0, \theta, \phi)$ is determined by the properties of the reflecting target and can be estimated using either radiative transfer simulations or empirically based on multiangular observations (ref). After the derivation of spectral flux $F(\lambda)$, the broadband flux F_{BB} can be estimated through a simple integration

$$F^{BB} = \int_0^{\infty} F(\lambda) d\lambda = \int_0^{\infty} \frac{\pi I(\lambda,\theta_0,\theta,\phi)}{R(\lambda,\theta_0,\theta,\phi)} d\lambda. \quad (3)$$

In practice, the above procedure is often modified or simplified as a result of instrument characteristics and algorithm design. For example, the direct measurement of CERES is the spectrally integrated radiance $I_{CERES}(\theta_0, \theta, \phi) = \int_{300\,nm}^{5000\,nm} I(\lambda, \theta_0, \theta, \phi) d\lambda$ with $300\,nm$ and $5000\,nm$, respectively, corresponding to the lower and higher end of CERES' spectral response function (SRF). The broadband SW flux F_{CERES} reported in the SSF product is derived based on

$$F^{BB}_{CERES} = \frac{\pi I_{CERES}(\theta_0,\theta,\phi)}{\overline{R}(\theta_0,\theta,\phi)} = \frac{\int_{300\,nm}^{5000\,nm} \pi I(\lambda,\theta_0,\theta,\phi) d\lambda}{\overline{R}_{CERES}(\theta_0,\theta,\phi)}, \quad (4)$$

where $\overline{R}_{CERES}(\theta_0, \theta, \phi)$ in the denominator can be considered as a weighted average of the spectral ADM anisotropic factors given by

$$\overline{R}\left(\theta_0, \theta, \phi\right) = \frac{\int_{300\,nm}^{5000\,nm} R\left(\lambda,\theta_0,\theta,\phi\right)F(\lambda)d\lambda}{\int_{300\,nm}^{5000\,nm} F(\lambda)d\lambda}. \quad (5)$$

In the CERES algorithm, $\overline{R}\left(\theta_0, \theta, \phi\right)$ is "scene type" dependent, where scene type is a combination of variables (e.g., surfacetype, cloud fraction, cloud optical depth, cloud phase, aerosol optical depth, precipitable water, lapse rate) that is used to group the data to develop distinct ADMs. In this study we will use the CERES ADM for the desert region developed by (Loeb 2005). More details can be found in section 3.2.

When we developed the algorithm to derive the TOA flux from the EMIT observations, an important consideration is whether our results compare well with the collocated CERES flux products. This motivated us to derive the broadband flux using a methodology similar to CERES' as follows

$$F_{EMIT}^{BB} = \frac{\pi I_{EMIT}\left(\theta_0,\theta,\phi\right)}{\overline{R}\left(\theta_0,\theta,\phi\right)} = \frac{\int_{314\,nm}^{2493\,nm} \pi I(\lambda,\theta_0,\theta,\phi)d\lambda}{\overline{R}_{CERES}\left(\theta_0,\theta,\phi\right)}. \quad (6)$$

The derivation of the spectral flux $F(\lambda)$ from EMIT's spectral radiance observation $I\left(\lambda, \theta_0, \theta, \phi\right)$ is based on the following simple equation:

$$F_{EMIT}(\lambda) = \frac{\pi I_{EMIT}\left(\lambda,\theta_0,\theta,\phi\right)}{\overline{R}\left(\theta_0,\theta,\phi\right)}, \quad (7)$$

3.1. **Spectral integration**:

Explain how to use PACE OCI to estimate the missing energy in the UV

Spectral region	Wavelength range	

UVB and shorter	< 314 nm	Neglected
UVA	314 nm to 381 nm	Empirical regression
Visible to SWIR	381 nm to 2493 nm	EMIT observation
SWIR and longer	>2493 nm	Neglected

Figure: show the regression analysis.

3.2. Angular distribution model

Explain how the broadband irradiance is converted to

Figure: polar contour of ADM of desert region from CERES

4. Validation and uncertainty analysis

4.1. Comparison of TOA flux with CERES

Figure: show collocation map

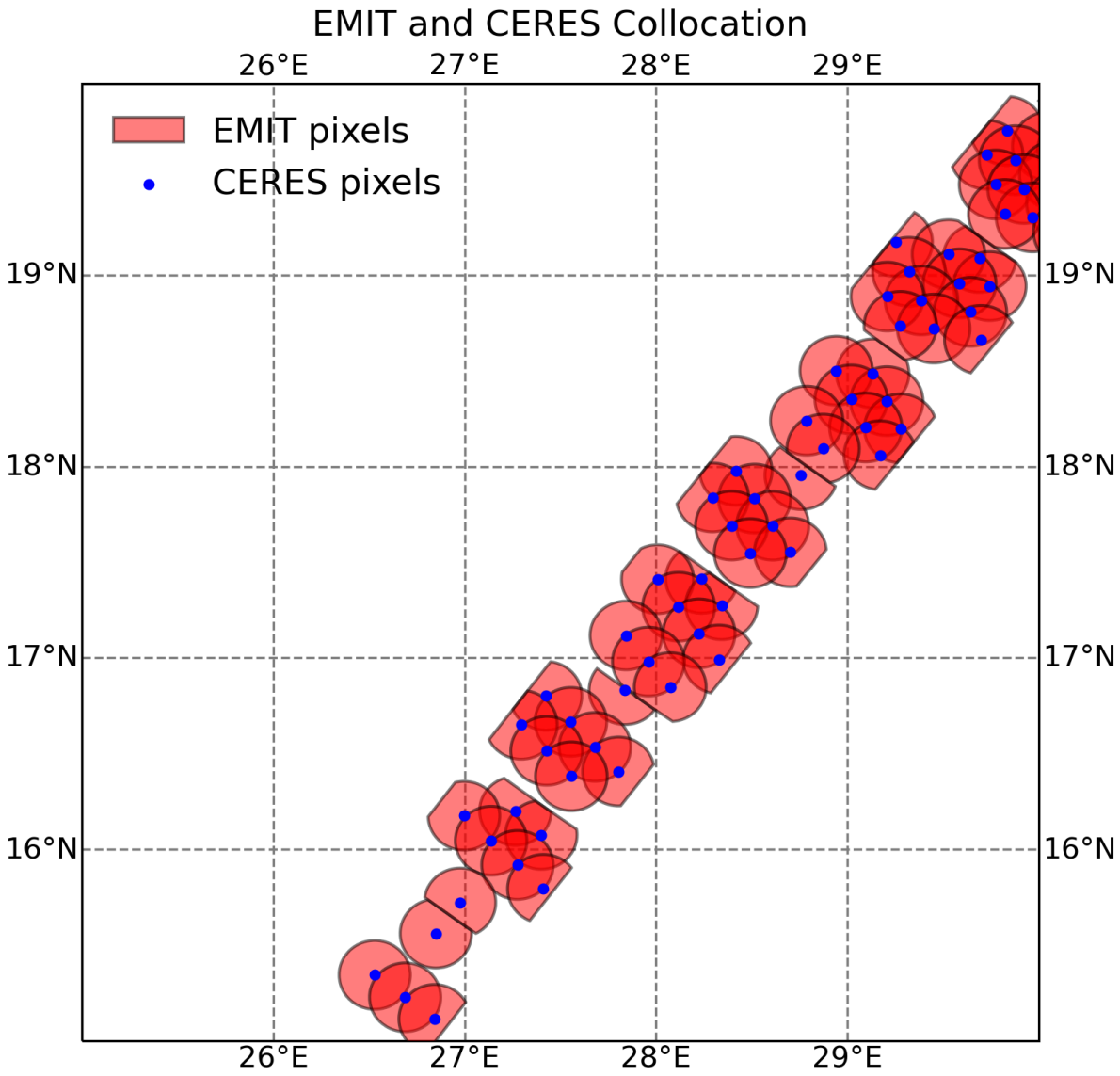
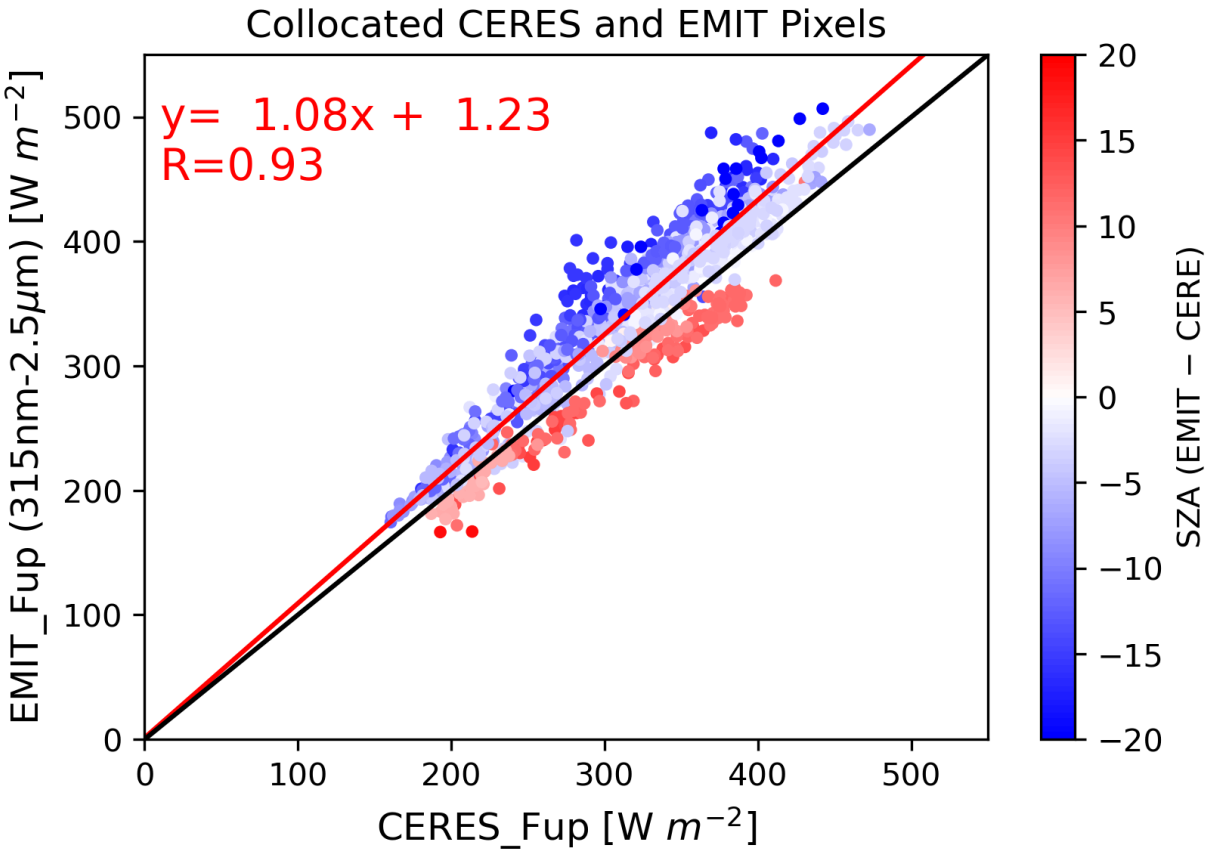
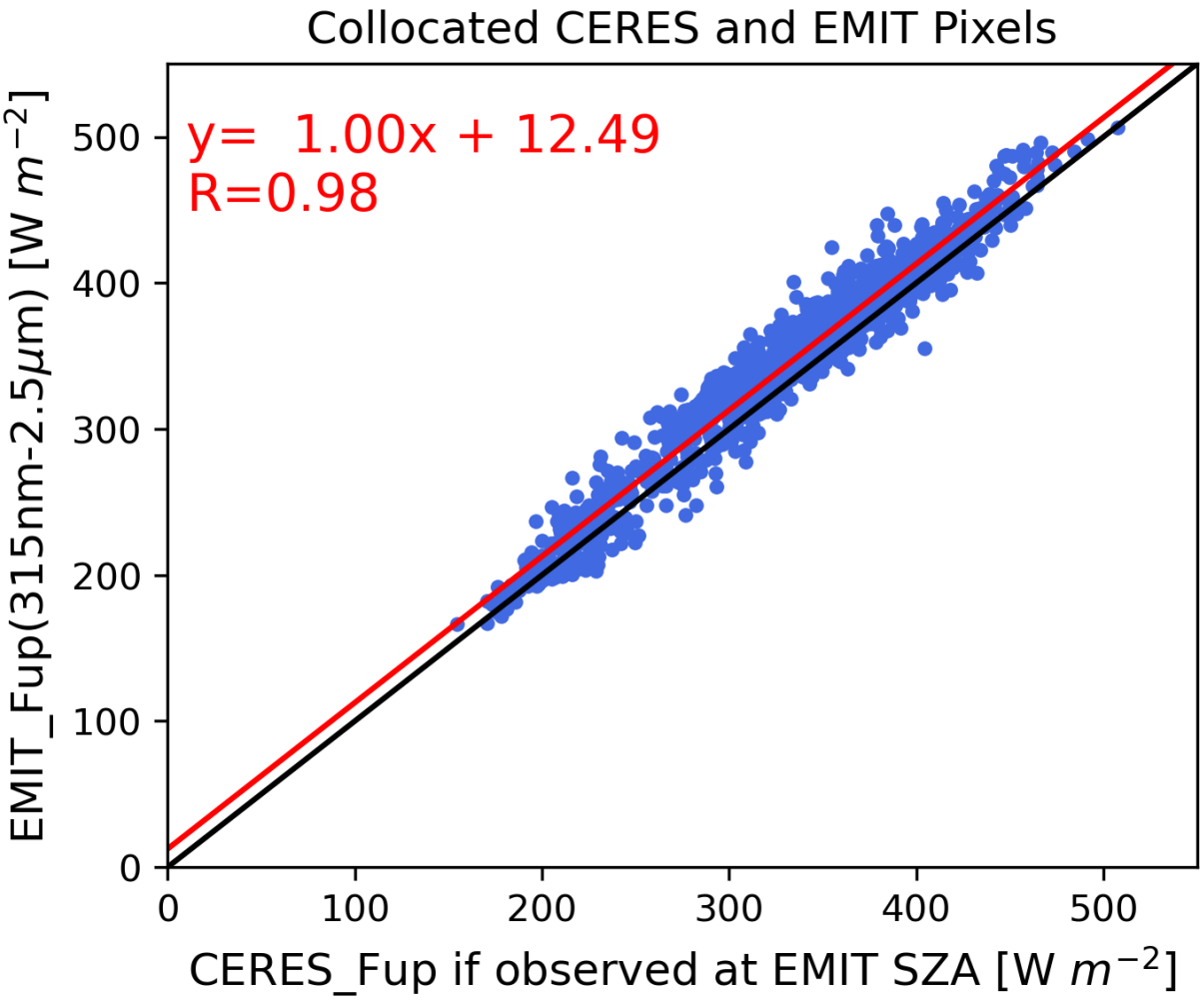


Figure: show regression of two fluxes. Raw data and solar angular adjusted.


Collocated CERES and EMIT Pixels
y= 1.08x + 1.23
R=0.93
EMIT_Fup (315nm-2.5μm) [W m⁻²]
CERES_Fup [W m⁻²]
SZA (EMIT − CERE)
0
100
200
300
400
500
20
15
10
5
0
−5
−10
−15
−20



4.2. Uncertainty due to missing energy

4.3. Uncertainty due to ADM

Figure: show spectrally resolved ADM for a case study.

Summary and Discussion

Is it possible to find a case to demonstrate the spectrally resolved flux? For example, it would be great to find a grid box, one day with almost no dust and the other day with significant dust. Then show the spectral difference between them.

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